Paleomagnetism of 1140 to 1150 Million-Year Diabase Sills from Gila County, Arizona[1]

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Diabase sills are abundantly developed in Gila County, central Arizona, and they are isotopically dated as 1140–1150 m.y. Stable magnetizations can be retrieved by AF or thermal demagnetization of the diabase. The over-all mean direction of magnetization of 98 specimens from 8 sites in the Globe–Salt River canyon area is at declination 285°, inclination 49° ($\alpha_{95} = 7°$, $k = 56$) with a corresponding paleomagnetic pole at 179°E, 27°N ($\delta p = 6°$, $\delta m = 9°$). This is identical with two published poles for diabase of the same age from the Sierra Ancha, Arizona. Furthermore, all three poles are in close agreement with those from igneous rocks of Keweenawan age exposed in the Lake Superior region. The stable Globe–Salt River canyon diabases have median Q' ratios near 0.8 and show no correlation between large amounts of secondary component removed during demagnetization and either low Q' or high susceptibility. The opaque minerals in the sills are highly oxidized titanomagnetites, unoxidized ilmenites, and a pyrite-pyrrhotite-chalcopyrite assemblage. The main carrier of the stable NRM appears to be a magnetite with very little titanium, produced during deuteric oxidation, and partly showing a single domain behavior.

An extensive diabase sill complex intrudes Younger Precambrian sedimentary rocks in central Arizona. Paleomagnetic studies on diabase from one part of the area, the Sierra Ancha, showed that the directions of natural remanent magnetization (NRM) had low dispersion and that this was further reduced by treating the specimens in alternating fields (AF) of up to 5000 oe [*Helsley*, 1965]. Furthermore, paleomagnetic poles computed for the Sierra Ancha diabase were in broad agreement with those from other rocks of the same general age [*Dubois*, 1964; *Helsley*, 1965].

We now report detailed studies on the same diabase exposed in the Globe and Salt River Canyon areas (Figure 1). A paleomagnetic result from these sills in Central Arizona is important to considerations of the Precambrian field, because the rocks are isotopically dated, and they are geographically distant from the Lake Superior region where the majority of North American, Late Precambrian paleomagnetic determinations of comparable age have been derived.

General Geology

Precambrian rocks in Arizona have been commonly divided into 'Younger' and 'Older' systems based on structural and lithologic grounds [*Butler and Wilson*, 1938]. The Younger Precambrian consists of non-metamorphosed sedimentary sequences which overlie Older Precambrian crystalline rocks with a marked structural and metamorphic unconformity. This usage has been followed in subsequent summaries of basement geology in Arizona [e.g., *Anderson*, 1951; *Wilson*, 1962].

Younger Precambrian rocks include the Troy quartzite and the Apache group; the latter consists of conglomerate, shale, quartzite, and an infrequent capping of vesicular basalt (Table 1). Both sedimentary sequences are extensively intruded by diabase sills which are up to 400

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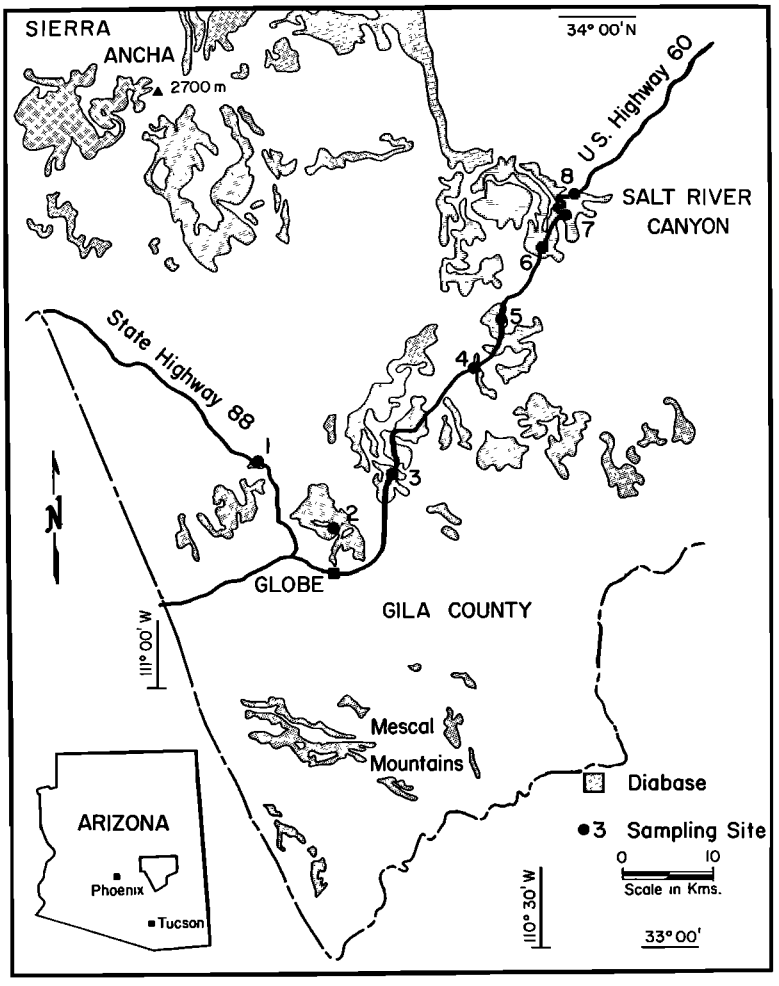


Fig. 1. Geologic map of Younger Precambrian diabase outcrops in Gila County, Arizona [after *Wilson et al.*, 1959], showing collecting sites.

m thick [*Shride*, 1967; *Smith*, 1970; *Nehru and Prinz*, 1970]. The sills frequently transgress several hundred feet of section in the Sierra Ancha. Differentiation is common within the large sills, and there is evidence for multiple intrusion at some localities [*Nehru and Prinz*, 1970]. Nevertheless, no regional differences have been found among the intrusives, so that they are considered to form one petrologically homogeneous system [*Neuerburg and Granger*, 1960] and therefore are very probably of the same age throughout central Arizona.

Isotopic Ages

The age of the Younger Precambrian diabase has been determined by two isotopic methods. The U/Pb method on cogenetic zircon systems from granophyric differentiates of the diabase yields discordant ages for zircon and uraninite, varying from 80 to 1100 m.y. [*Silver*, 1960, 1963]. When these data are plotted on a 'Concordia' diagram, an upper intersection indicates an original age of 1150 ± 30 m.y. and a lower intersection suggests a disturbing event at 80 m.y. This age compares with Pb 207/206 ages of 930 to 1104 m.y. reported for uraninite and galena in the Dripping Spring quartzite by *Neuerburg and Granger* [1960]. Neuerburg and Granger argue that the uranium deposits must be related genetically to the diabase because of

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TABLE 1. Schematic Stratigraphic Section through the Younger Precambrian of Central Arizona [after *Wilson*, 1962]

CAMBRIAN	
----------unconformity----------	
Diabase	
Troy quartzite	
----------unconformity----------	
Basalt	
Mescal limestone	
Dripping Spring quartzite	
	Apache group
Barnes conglomerate	
Pioneer shale	
Scanlan conglomerate	
----------unconformity----------	
OLDER PRECAMBRIAN	

their spatial relationship and simply because the diabase is the only known source of energy in the region to initiate mineralization.

A K/Ar apparent age of 1140 ± 40 m.y. was found for biotite from diabase in the Sierra Ancha [*Damon et al.*, 1962], which is consistent with the U/Pb ages. More important, the biotite age shows no effect of the possible reheating episode that is implied by the lower intersection of the Concordia plot. Thus, an age of 1140 to 1150 m.y. is well established for the diabase sills of central Arizona [*Livingston and Damon*, 1968; *Damon*, 1968].

SAMPLING DETAILS

In Gila County the diabase is exposed in three main areas, and this study concerns outcrops in the Globe–Salt River canyon region (Figure 1). We did not collect from outcrops in the Mescal Mountains because they are structurally complicated and highly weathered. Eight sites are represented in the collection (Table 2). At seven of the sites, from 3 to 24 oriented cores (samples) were obtained, each 2.5 cm in diameter and at least 8 cm long, by means of a gasoline powered drill [*Helsley*, 1967*a*]. Oriented block samples were collected at site 7 on the south rim of Salt River canyon, to compare the relative within-site orientation accuracies of block sampling and field drilling. Azimuthal orientation for both methods was by magnetic compass.

Fresh samples of the diabase can readily be obtained in this area, since there are numerous deep road cuts. No samples were drilled in hornfelsic and fine grained margins of the sills (because the contact was not always well defined) nor in the country rock sediments (because they were generally brittle). Structural control on the attitude of the diabase is good, and samples were taken only where the attitude of the overlying sediments was fairly well defined as horizontal and the diabase was clearly a sill. Thus no structural correction is necessary for the samples.

The Younger Precambrian is not exposed continuously for long distances. The outcrops are separated by Quaternary gravels and basalts and Tertiary dacites, and by Older Precambrian crystalline rocks. This situation and the transgressive nature of the diabase make it unlikely that the same sill has always been sampled. In fact we believe that each site represents a different sill, and we argue that a representative collection has been made over the diabase system in the Globe–Salt River canyon area.

EXPERIMENTAL METHODS

Each field drilled core was sliced into two specimens 2.5 cm long; two cores were drilled in the laboratory from each block sample collected at site 7. The direction and moment of the NRM of the specimens were measured on one of two spinner magnetometers. One was a portable air-turbine spinner [*Helsley*, 1967*b*] that had a minimum operating sensitivity in the laboratory of about 2×10^{-7} emu/cc at which the phase reading was reliable to 5°. The other was a PAR spinner, model SM-1 with a phase sensitivity of ±0.5° and a magnitude sensitivity of better than 1% at an operating sensitivity near 1×10^{-7} emu/cc.

The AF demagnetization equipment used in the study is described by *Helsley* [1969] and enables maximum fields of 2800 oe to be pro-

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TABLE 2. Sampling Details

Site	Location	Country Rock*	No. Samples	Remarks
1: Salt River Mountains, 16 km NW of Globe, State Hwy. 88	33.5°N, 100.9°W	Apache group sediments	6	Rather weathered outcrops along canyon walls
2: Globe hills, 5 km north of Globe, State Hwy. 88	33.5°N, 110.8°W	Apache group sediments	6	Isolated Precambrian outcrops surrounded by Quaternary gravels and Tertiary dacites
3: Apache Mountains, 5 km south of McMillanville ghost town, U.S. Hwy. 60	33.5°N, 110.7°W	Dripping Spring quartzite	6	Fresh road cuts
4: Jackson butte, 35 km north of Globe, U.S. Hwy. 60	33.7°N, 110.6°W	Precambrian granite	6	Contact horizontal; road cut drilled 20 meters below contact
5: Timber Camp Mountains, 48 km north of Globe, U.S. Hwy. 60	33.7°N, 110.6°W	Pioneer shale	5	Scarp within pine forest
6: Mule Hoof Bend canyon, 2 km south of Seneca, U.S. Hwy. 60	33.7°N, 110.5°W	Troy quartzite	3	Fresh road cut
7: Salt River canyon, south rim, U.S. Hwy. 60	33.8°N, 110.5°W	Mescal limestone	6	7-meter sill in roadcut; block samples along vertical profile
8: Salt River canyon, north rim, U.S. Hwy. 60	33.8°N, 110.5°W	Mescal limestone	24	Samples throughout a sill 150 meters thick

*After *Shride* [1967].

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duced by a 400 cps generator. Precautions were taken to ensure that no ARM's were induced during demagnetization by tumbling the specimen, by repeating the demagnetization process with the specimen inverted in the holder, and most importantly, by having the demagnetizing coils located inside a walk-in 3.1-m mu-metal room with a residual field of less than 200 gammas.

Thermal demagnetization was carried out by the stepwise method in a non-inductively wound electric furnace [*Helsley*, 1969] that is surrounded by two sets of Helmholtz coils in the shielded room. Twelve specimens can be accommodated during any one heating cycle and the temperatures on each of four shelves are monitored by platinum thermocouples. The acquisition of partial TRM's was minimized by cooling the specimens rapidly and in a residual field of less than ± 5 gammas.

One of the two laboratory cleaning techniques was applied to all the samples from sites 1 through 7 as a pilot procedure, in order to find the over-all temperature or field that would most effectively clean a second specimen from each sample. To get some idea of the relative efficiency of either method in removing unwanted magnetization, the collection of 24 samples from site 8 was arbitrarily divided into two equal groups. One set was AF demagnetized and the other was thermally demagnetized; only one specimen from each independently oriented drill core at this site was used in the magnetic analysis.

Site mean directions for each demagnetization stage were computed according to *Fisher* [1953]. This gives R, the resultant of N unit specimen vectors, and $\mathbf{k}$, an inverse measure of the dispersion. *Irving et al.* [1961] showed that a value $\mathbf{k_0}$ could be calculated for a statistically random population and that this condition is true if $\mathbf{k/k_0} \leq 1$. This is a useful method of normalizing the dispersions among several populations with different numbers of specimens.

In this study we will assume that the primary magnetization has been recovered when (1) there is no longer any systematic movement of the mean NRM direction for a site, and (2) the dispersion of the individual directions is significantly minimal (that is, they have become tightly grouped). We can demonstrate this last condition statistically by using the tabulations of *McElhinny* [1964] to compare $\mathbf{k}$ values after successive demagnetization steps, to test whether there has been a significant convergence of the NRM directions. If the two assumed conditions coincide, there is strong evidence that the demagnetization process has been effective in removing secondary magnetizations.

THERMAL DEMAGNETIZATION

One specimen from each sample at sites 1, 2, 6, and 7 and the twelve specimens from site 8 were heated successively to 200°, 400°, 500°, 530°, and 560°C. In some cases the 500° or 530°C stages were omitted when there was little direction change after the previous step. The specimens from site 8 were subjected to a further step at 600°C, but since their intensities had decreased by 10^3 and the dispersion of the NRM directions increased markedly, no experiments were made at this temperature for the other sites.

The precision ratio $\mathbf{k/k_0}$ at each temperature for each site is shown in Figure 2. The temperature ordinate is not drawn to scale because we do not wish to imply that the precision ratio can necessarily be interpolated between temperatures. The progressive demagnetization paths representing the site mean directions are shown in Figure 3.

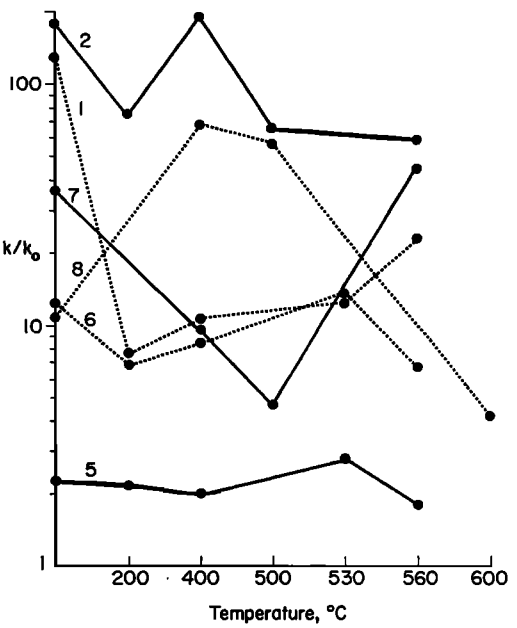


Fig. 2. Change of precision in terms of $\mathbf{k/k_0}$ (plotted on a log scale) for each site (numbered) during thermal demagnetization.

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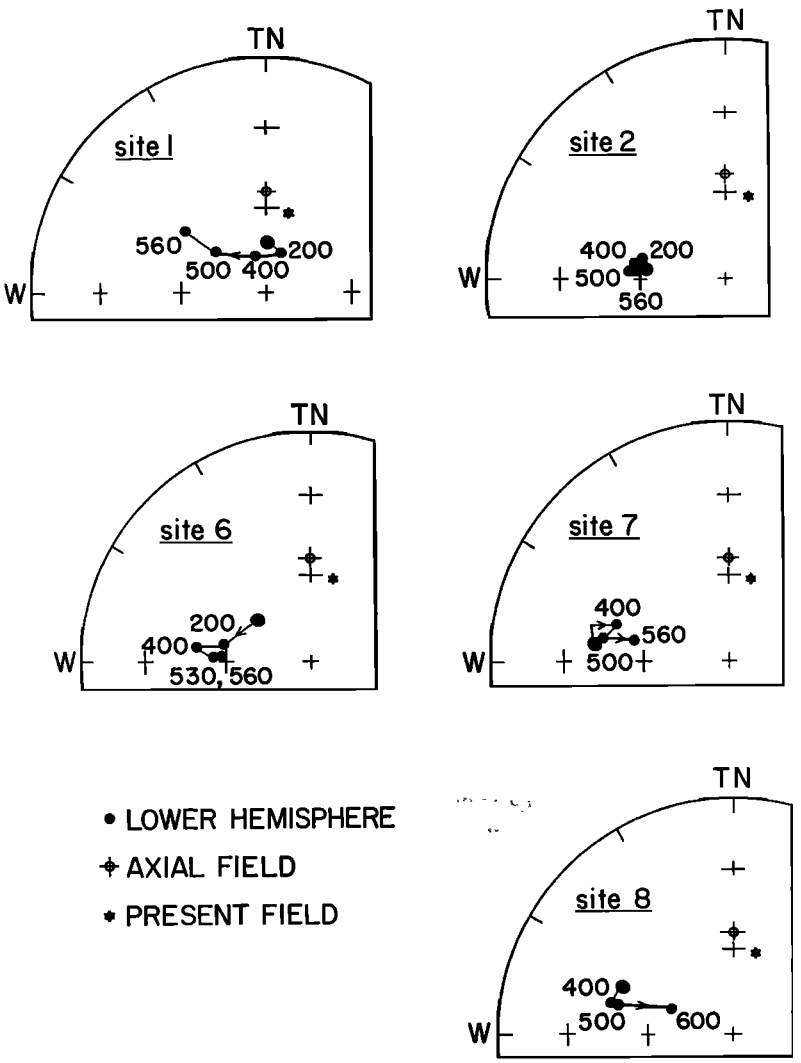


Fig. 3. Progressive thermal demagnetization paths for sites 1, 2, 6, 7, and 8, plotted on an equal area projection. Temperatures are in degrees centigrade. Axial and present (inclined) dipole fields are also shown.

Data from site 5 are peculiar and are not plotted; they are discussed in a later section. The dispersion remains poor (Figure 2) and the moments of all specimens from this site are much greater than those from other sites. We will assume tentatively that the site has been struck by lightning.

The only systematic vector rotation observed occurs at site 1 and to a less extent at site 6. At site 1 the fact that the mean vector is still moving at 560°C implies that the end point may not have been reached (Figure 3) and that we should have continued demagnetization. This is supported by the thermal decay curves from this site, because in two of the specimens over 20% of the initial NRM remains at 560°C, whereas in specimens from other sites at 560°C less than 5% remains. Nevertheless, the dispersion is minimal (i.e., $\mathbf{k}/\mathbf{k}_0$ is at maximum) at this temperature so it is reasonable to select 560°C as an effective demagnetization temperature for site 1. It is interesting that the $\mathbf{k}$ value for the initial NRM directions at this site is an order of magnitude higher than it is at any other

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temperature (Figure 2). It implies that a large secondary component is present in the initial NRM, and this is confirmed by the initial direction being close to the present field.

At site 6 there is systematic vector rotation to 400°C, but this is not the temperature of minimum dispersion. The maximum $\mathbf{k/k_0}$ occurs at 530°C and this temperature was chosen as the demagnetization temperature for this site. Nevertheless the dispersion at this temperature is not significantly different (at the 95% level) from that at 400° or 560°C. At sites 2, 7, and 8 there is no systematic vector rotation, although the mean direction changes slightly after each demagnetization stage. In all cases the effective cleaning temperature was selected as the stage at which minimum dispersion occurred.

At site 7 the initial NRM directions are the most tightly grouped and this low dispersion is not retrieved until 560°C, which was the temperature chosen for over-all demagnetization. At site 2 we again observe that the dispersion in the initial NRM directions is very low, although by inspection they are not related to the present field direction. In any case, there is little directional change during thermal demagnetization and the least dispersion occurs at 400°C.

Thermal decay curves drawn for each site are typically knee shaped and demonstrate that most of the natural remanence is carried by a thermally discrete component [*Irving and Opdyke*, 1965] with a Curie temperature well in excess of 500°C.

AF Demagnetization

One specimen from each sample at sites 3 and 4, and twelve specimens from site 8 were subjected to AF demagnetization studies in successive fields of 88, 350, 700, and 1400 oe. Site means were computed as before, and the precision at each AF stage in terms of $\mathbf{k/k_0}$ is shown in Figure 4. The demagnetization paths, represented by the mean directions for each site (except site 5), are given in Figure 5.

There is systematic vector rotation at sites 3 and 4. Calculation of the vectors removed from site 3 specimens shows that they reflect the present field; those removed from site 4 specimens suggest (by comparison with Upper Cretaceous data from the Sierra Nevada [*Currie et al.*, 1963] and Utah [*Kilbourne*, 1969]) that this site was slightly remagnetized by heating associated with Laramide events. (Recall the 80 m.y. event on the Concordia plot of *Silver* [1960, 1963].) The dispersion is minimal at 1400 oe for sites 3 and 4, and this field was selected for demagnetization of the remaining samples from this site. There is very little change in direction at site 8 and the dispersion is minimal at 88 oe.

The normalized AF decay curves for two sites are plotted in Figure 6. Components of high coercivity are present in the specimens from site 4, whereas the site 8 specimens have distributed coercivities. It is rather interesting that the high coercivity at site 4 is associated with about 40° of vector rotation (and possible Laramide remagnetization), whereas the broader spectrum of coercivities seen at site 8 is associated with negligible direction change.

Site 5 Behavior

The remanent magnetism at site 5 is anomalous and suggests that it has been struck by lightning.

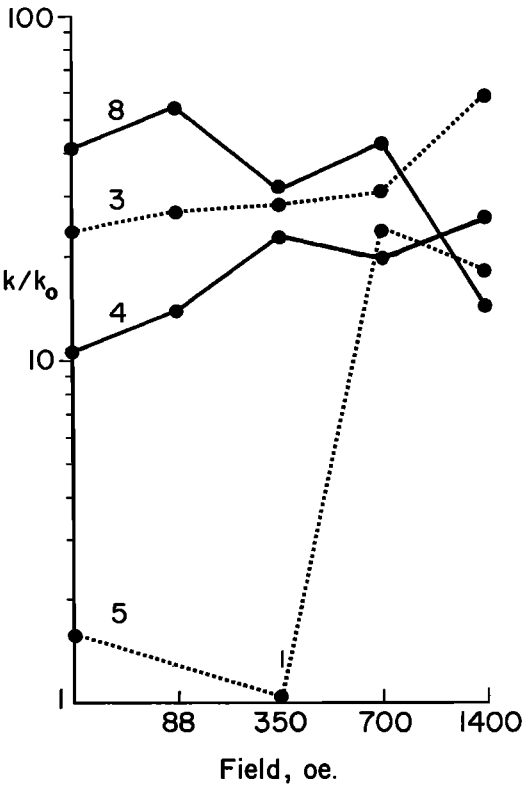


Fig. 4. Change of precision in terms of $\mathbf{k/k_0}$ (plotted on a log scale) for each site (numbered) during AF demagnetization.

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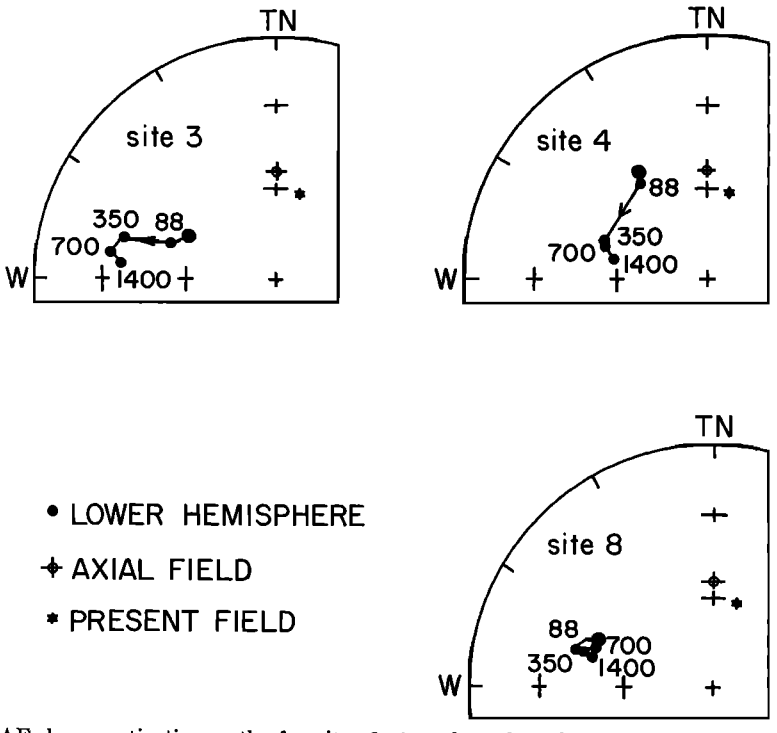


Fig. 5. AF demagnetization paths for sites 3, 4, and 8, plotted on an equal area projection. Fields in oersteds. Axial and present (inclined) dipole fields are also shown.

The intensities of specimens from this site are at least a factor 5 greater than those from other sites. Thermal demagnetization to 560°C has little effect. The mean tends to move toward the end points for other sites, although the $\mathbf{k}/\mathbf{k}_0$ factor is never greater than 3 (Figure 4). There is a steep decrease in the thermal decay curves at about 400°C that is very similar to one observed in lightning-affected samples from the Umkondo dolerite of southern Rhodesia [*McElhinny and Opdyke*, 1964]. It is unlikely that the primary remanence of site 5 specimens

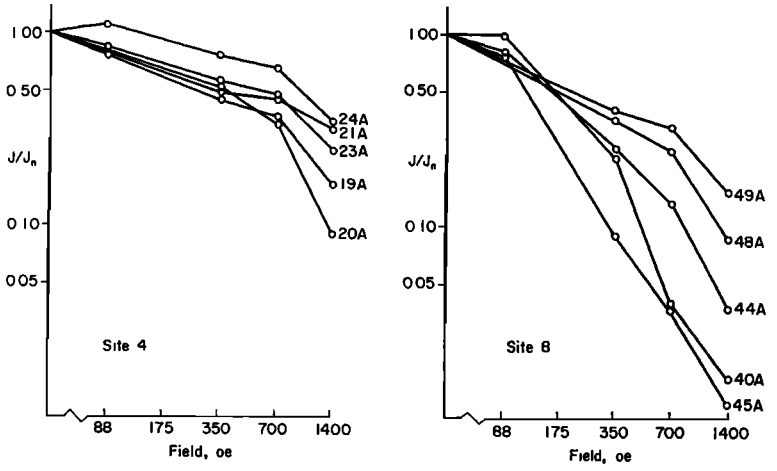


Fig. 6. Normalized AF decay curves for specimens from sites 4 and 8.

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lies in a carrier with a Curie point higher than 560°C, because the NRM decay curves indicate that less than 1% of the initial moment remains at this temperature.

Progressive AF demagnetization to 700 oe appears to completely remove the secondary components (Figure 7), and the NRM directions are then similar to those from other sites. The precision ratio (k/k_0) increases from essentially that for a random state initially and at 350 oe, to values at 700 and 1400 oe which compare favorably with those from other sites (Figure 4). Since the mean at 700 oe agrees well with other site means, it was felt that this field would effectively remove the unwanted IRM.

The curious fact is that all the samples from the site have been affected by the lightning stroke. The outcrop is a scarp face dipping at about 60° to the west in a forest clearing. Sampling was random over an area of about 25 m². This implies that the lightning bolt was an oblique strike near the crest of the scarp and that numerous line currents flowed down the scarp face. The resulting IRM is thus approximately normal to the plane of the scarp. It is rather surprising that all the initial directions are upward seeking. It may well be a coincidence from the sampling procedure, for both polarities might have been expected.

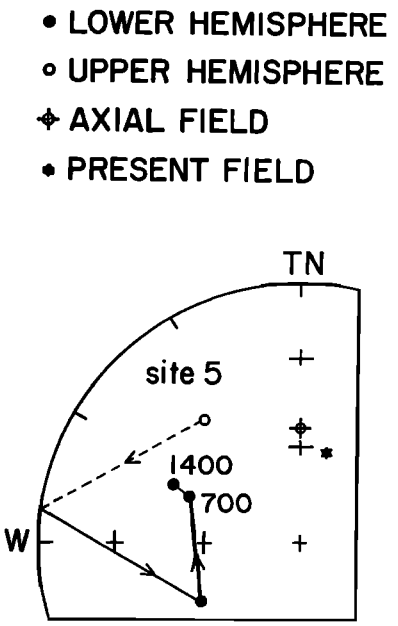


Fig. 7. AF demagnetization path for specimens from site 5, plotted on an equal area projection.

OPAQUE MINERALOGY

Specimens from all the sites show a common opaque mineralogy in having abundant oxidized titanomagnetites, unoxidized ilmenites, and variable amounts of a pyrite-pyrrhotite-chalcopyrite assemblage.

The titanomagnetic grains are generally equidimensional with corroded margins. The size range is from a few to 150 microns. Large grains are the most oxidized. The products are thin blades of ilmenite along [111] planes, and in many cases the host titanomagnetite contains spinel rods along [100] planes. According to the classification of *Ade-Hall et al.* [1968], this indicates high deuteric oxidation, in the 'Class 5 state.' However, there is no development of hematite, rutile, or pseudo brookite, and further, spinel rods also appear in the ilmenite lamellae. It suggests that this is perhaps an intermediate stage between the Class 4 and 5 states, where spinel unmixes out of the ilmenite lamellae before they break down to typical Class 5 oxidation products. This feature may be due to the reducing action of the sulphides, which are not considered in the usual oxidation classification of lavas.

Some of the titanomagnetite appears to contain ulvospinel. It was tentatively identified by its cloth texture [*Vincent*, 1960] in larger oxidized grains. The coarse grained diabases from sites 4 and 8 contain olivine, which is replaced by an assemblage of tentacle-like titanomagnetite and serpentine. Discrete ilmenite has a similar size range to titanomagnetite, but is not oxidized.

There is a great deal of pyrrhotite in many of the specimens (up to 40% by volume of total opaques in some cases), and it is generally intergrown with pyrite and chalcopyrite. The pyrrhotite is cubic, ranging in size from a few microns to about 50 microns.

CURIE POINTS

Curie points were determined on specimens from 3 sites in a field of about 1500 oe; two of these specimens contained abundant pyrrhotite. The curves are shown in Figure 8, and although all are reversible, there is a suggestion in two of the curves that a point of inflection occurs between 250° and 350°C. Since the Curie point of

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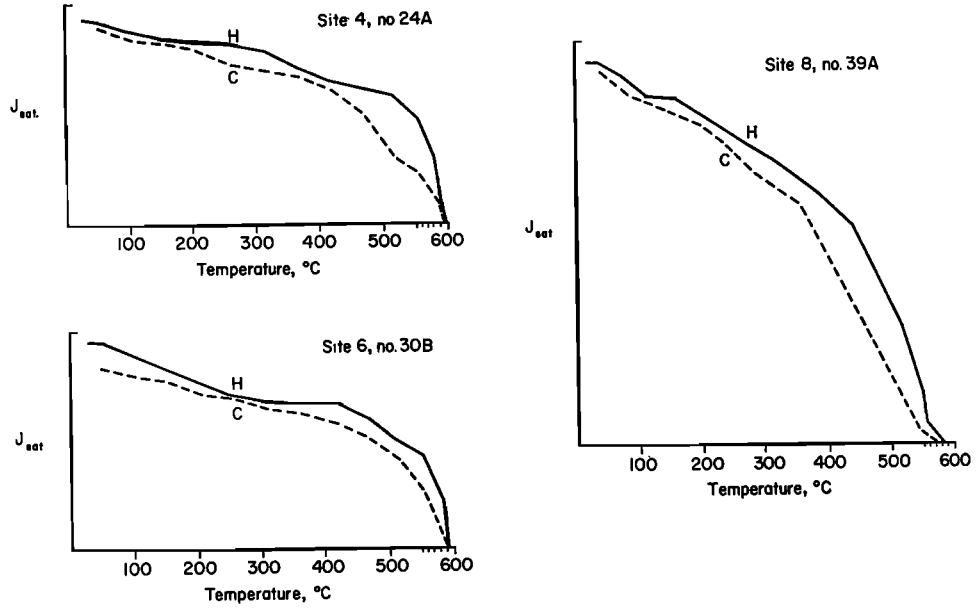


Fig. 8. Curie point determinations in a field of 1500 oe. Saturation magnetization scale is arbitrary. The specimens from sites 4 and 6 contain abundant pyrrhotite microscopically.

pyrhotite is 310°C [*Schwarz*, 1967], it suggests that this mineral contributes to the saturation magnetization. Nevertheless, we find it odd that pyrrhotite apparently contributes little to the NRM characteristics of the diabase.

The abrupt Curie points in the range 580° to 590°C for all the specimens are intriguing in view of the abundant titanium observed microscopically in the host spinel lattice. It implies that the carrier of the saturation magnetization is in fact a titanium free magnetite. It is possible that the point of inflection observed around 300°C is due to the titanium-rich host titanomagnetite. However, whereas this phase is present in all the specimens examined microscopically, the point of inflection is not observed in every Curie point determination. Thus we can only tentatively suggest that the main carrier of the saturation magnetization is also likely to be the carrier of the stable remanence. If this is a pure magnetite, it implies that it was produced during the deuteric oxidation process; the stable NRM may be a composite of TRM and high temperature CRM as *Grommé et al.* [1969] have observed in Hawaiian lavas. This is a further example of the remark by *Graham* [1953] and discussed by, among others, *Ade-Hall* [1969] and *Larson et al.* [1969], that effective rather than over-all grain size is of ultimate importance to magnetic stability.

Thus we can argue that the main carrier of the stable NRM in these Arizona diabases is a magnetite that contains very little titanium (because of its high Curie point), that is produced by the deuteric oxidation process (because the host oxide is titanium rich), that is of variable coercive force (because of the coercively distributed but highly stable magnetization observed at site 8), and that possesses, in part, single-domain behavior (because of its stability up to 1400 oe and the small grain size of the magnetite produced by deuteric oxidation).

ARTIFICIAL THERMOREMANENCE

A TRM was given to three specimens from different sites by cooling them from 750°C in a field of 0.3 oe. The TRM directions were within 3° of the axis of the applied field, and this is evidence that the specimens are magnetically isotropic. Two of the specimens were thermally demagnetized and their decay curves are identical with their NRM decay curves; this suggests that the NRM is a true TRM.

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INTENSITIES, SUSCEPTIBILITIES, AND Q' RATIOS

The range in intensity and susceptibility and their ratio (Q') for specimens from each site are given in Table 3; medians are quoted because they will be less influenced by extreme values. Susceptibility measurements were made only on those specimens used in the pilot demagnetization.

The over-all susceptibility range is small; the median value is 3.9×10^{-3} emu/oe/cc with a range of 1.6 to 10.2×10^{-3} emu/oe/cc. It suggests that the magnetic mineral content is approximately uniform from specimen to specimen. In contrast, there is a wide range in intensity. It varies from 1.0 to 35.1×10^{-3} emu/cc with a median at 3.6×10^{-3} emu/cc. Extreme values are found at site 5, where the intensity is a factor of 5 higher than at other sites. Yet the susceptibility range at this site is similar to that from other sites, so the idea that site 5 has been struck by lightning is supported.

The Q' range is from 0.34 to 11.90 with a median at 0.84. The high values are mainly from site 5, and they bear out the observation by *Strangway* [1965] that anomalously high Q' values are typical of rocks struck by lightning.

There appears to be little correlation of either of these three parameters with stability of NRM, in terms of secondary components removed during demagnetization. The lowest median Q' ratio is at site 3, and there the direction change is about the same as at site 4, which has the second highest median Q' ratio (disregarding site 5, which is an exception).

OVER-ALL DEMAGNETIZATION

Duplicate specimens from all the samples at sites 1 through 7 and the remaining specimens from site 5 were subjected to demagnetization according to the arguments outlined in an earlier section. Site means were computed according to *Fisher* [1953]; they are plotted with the associated circles of 95% level confidence in Figure 9 and listed in Table 4.

An F test [*Watson*, 1956] shows that the two mean directions obtained by AF and thermal demagnetization of site 8 specimens are not significantly different. The over-all mean for this site is at declination 285°, inclination 44° ($\alpha_{95} = 7°$, $\mathbf{k} = 1099$) with a corresponding pole at 174°E, 25°N ($\delta p = 6°$, $\delta m = 9°$). The site represents a thick (150 meter) sill. The over-all mean is approximately central to the other site means; this pattern suggests that secular variation effects have been averaged out by sampling systematically through the sill.

Demagnetization procedures reduce the initial NRM dispersion at all the sites. While this is not always significant at the 95% confidence

TABLE 3. Range and Median Values of Intensity, Susceptibility, and Q' Ratio for the Diabase

Site	Intensity, x 10^{-3} emu/cm^3*		Susceptibility, x 10^{-3} emu/oe/cm^3†		Q'	
	Range	Median	Range	Median	Range	Median
1	1.5 to 35.1	3.3	2.9 to 6.3	6.0	0.54 to 5.60	0.60
2	3.1 to 3.9	3.6	1.9 to 2.9	2.6	1.27 to 1.93	1.47
3	1.5 to 2.5	2.0	5.0 to 5.9	5.5	0.34 to 0.42	0.39
4	1.7 to 3.0	2.6	2.0 to 2.5	2.3	0.83 to 1.32	1.15
5	8.9 to 28.5	22.7	2.1 to 2.6	2.3	5.33 to 11.90	9.44
6	1.8 to 6.0	5.2	3.0 to 4.8	4.1	0.38 to 1.91	0.53
7	2.2 to 6.3	4.5	2.8 to 8.0	5.5	0.72 to 1.53	0.83
8(AF)	1.1 to 13.6	2.6	1.6 to 10.2	3.6	0.74 to 1.40	0.85

*All specimens used in study.
†Specimens in pilot demagnetization only.

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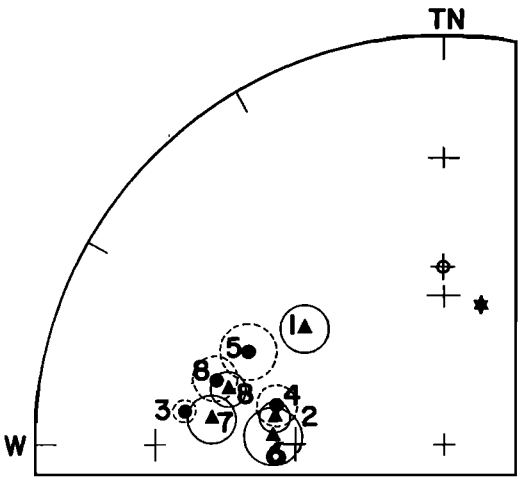


Fig. 9. Site mean directions of magnetization with circles of 95% confidence, plotted on an equal area projection. Thermal means are triangles with solid line circles of confidence; AF means are solid dots with dotted circles of confidence. Axial and present fields are also shown.

level, the site means are well grouped, and application of Hartley's maximum F ratio test [*Watson*, 1956] on the **k** values indicates that the precision is uniform among all the populations, using the five thermal means, the four AF means, or all nine means. Since the areal spread of the collecting sites is small, we have computed a grand mean simply by combining the eight site means (including the single mean for site 8); this is shown at the bottom of Table 4.

We can see that both demagnetization procedures have been equally effective in removing secondary moments. Where lightning has affected the remanence, AF demagnetization is more effective in removing the induced IRM. Site 7 was block sampled, and since the dispersion is similar to that from other sites, in this particular case there has been no loss of precision during laboratory reorientation of the blocks.

PALEOMAGNETIC POLES

Paleomagnetic poles corresponding to the eight site mean directions of magnetization are

TABLE 4. Site Mean Directions of Magnetization and Paleomagnetic Poles

Site No.	Temp., °C	Field, oe	Dec.	Inc.	N	R	k	α_{95}	Location	δp	δm
					Thermal Demagnetization						
1	0		11	66	12	11.892	102	4	72°N, 86°W	5	7
1	560		360	53	12	11.895	105	4	48°N, 174°E	4	5
2	0		279	58	12	11.929	156	3	26°N, 169°W	3	5
2	400		279	55	12	11.955	242	2	25°N, 172°W	2	4
6	0		304	64	6	5.942	87	7	46°N, 165°W	9	11
6	530		273	55	6	5.951	101	6	21°N, 169°W	6	9
7	0		279	54	12	11.826	63	5	25°N, 173°W	5	7
7	560		276	42	12	11.887	97	4	17°N, 177°E	3	5
8	0		293	47	12	11.501	22	9	33°N, 173°E	7	12
8	400		285	45	12	11.917	133	3	25°N, 175°E	3	4
					AF Demagnetization						
3		0	297	57	12	11.874	86	4	39°N, 176°W	5	6
3		1400	278	36	12	11.948	211	2	16°N, 173°E	2	3
4		0	327	41	12	11.601	27	8	59°N, 150°E	6	10
4		1400	284	55	12	11.887	98	4	28°N, 174°W	4	6
5		0	327	39	8	6.962	7	23	25°N, 103°E	16	27
5		700	296	46	8	7.925	93	5	35°N, 171°E	4	7
8		0	292	48	12	11.839	68	5	32°N, 175°E	4	6
8		88	286	41	12	11.879	91	4	25°N, 172°E	3	5
					Over-all Mean for Sites 1 through 8						
			285	49	8	7.876	56	7	27°N, 179°E	6	9

Field is alternating field; Dec. is declination of the north-seeking magnetic vector, east of true north; Inc. is its inclination, positive downward; N is the number of specimens; R is the vector sum of N unit vectors; $k = (N - 1)/(N - R)$; α_{95} is the semi-angle of the cone of 95% level confidence; Location is position of the paleomagnetic pole; δp and δm are the semi-minor and semi-major axes of the oval of 95% level confidence.

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plotted in Figure 10, with the over-all mean and the two poles for diabase from the Sierra Ancha quoted by *Dubois* [1964] and *Helsley* [1965]. There is excellent agreement among the three studies, indicating that there are no large age differences among the Precambrian sills in Central Arizona and suggesting that no large secular variation effects are present.

The 1140–1150 m.y. poles reported in this study fall within the locus of poles quoted by *DuBois* [1962], *Helsley* [1965], *Beck and Lindsley* [1969], *Beck* [1970], *Palmer* [1970], and others for Middle Keweenawan igneous rocks (1000–1200 m.y.) exposed around the Lake Superior region. In particular, *Beck* [1970] refers to a pole at 168°W, 34.5°N for the Duluth gabbro complex (*D* in Figure 10), which has been isotopically dated as 1115 m.y. by Rb/Sr methods [*Faure et al.*, 1969]; *Palmer* [1970] summarizes a pole at 177°W, 29°N for nine normally magnetized Keweenawan igneous rocks (*K* in Figure 10). The excellent agreement between the Arizona and Lake Superior poles is evidence that the field was dipolar in the period 1115 to 1150 m.y., because rocks of essentially the same age but from different parts of the same continent (that is, from different magnetic latitudes) give a common paleomagnetic axis. If this dipolar model for the paleo field is correct, it further implies that there has been no relative displacement between the two areas greater than about 1000 km since 1115 to 1150 m.y. This means that these parts of the North American craton have been a single crustal plate since about 1150 m.y.

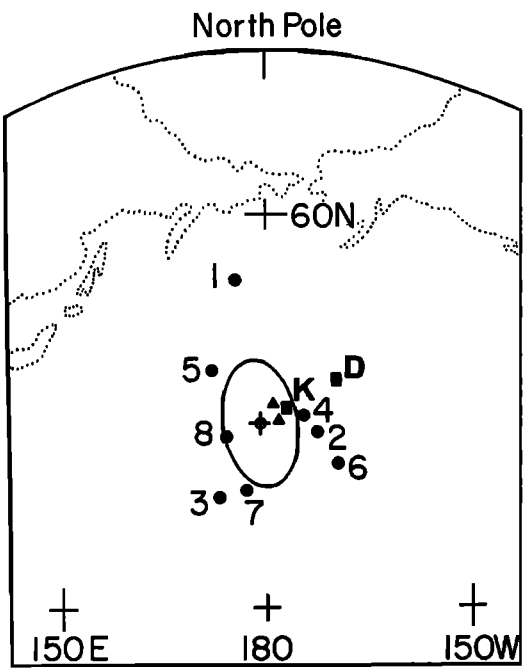


Fig. 10. Paleomagnetic poles for each site mean, plotted on an equal area projection. The crossed circle symbol is the over-all mean for the Globe–Salt River canyon diabase, with its oval of 95% confidence. Triangles are the poles quoted by *Dubois* [1964] and *Helsley* [1965] for diabase from the Sierra Ancha, Arizona. The square marked *K* is a mean for Keweenawan igneous rocks of normal polarity [*Palmer*, 1970] and the square marked *D* is a mean for the Duluth gabbro complex [*Beck*, 1970].

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